Cardiac Electrical Dipole Projections: 12-Lead Standard vs. Ambulatory Wearables

Physical spatial vectors dictate what information is observable vs. mathematically ambiguous

A. Hospital 12-Lead Standard

Full 3D Heart View

Electrodes:
10 physical skin electrodes

Spatial Vectors:
12 leads (6 limb, 6 precordial)

Dipole Coverage:
Complete 3D Space (X, Y, Z)

Diagnostic Reach:
STEMI, NSTEMI, LVH, QTc

Clinical Reality:
Confined to hospital carts

B. Smartwatch (Lead I)

Horizontal 1D Projection

Electrodes:
2 dry metal contacts (wrists)

Spatial Vector:
Single horizontal axis (X)

Dipole Coverage:
1D transverse projection only

Diagnostic Reach:
Atrial Fibrillation, HR, Rhythm

Blind Spot:
Blind to vertical & posterior

C. Multi-Channel Patch

Planar 2D Projection

Examples:
CardioSTAT, Zio XT, Icentia

Spatial Vectors:
2 dual-channel vectors (X, Y)

Dipole Coverage:
Planar 2D frontal coverage

Diagnostic Reach:
Complex arrhythmia, flutters

Blind Spot:
Limited anterior chest depth

D. 3-Lead Triad (I, II, V2)

Orthogonal 3D Basis

Electrodes:
3-wire clinical configuration

Spatial Vectors:
X: Lead I, Y: Lead II, Z: V2

Dipole Coverage:
Orthogonal 3D Basis (X, Y, Z)

Diagnostic Reach:
Near-lossless 12L recovery

Clinical Power:
In-clinic triage with 12L power

Key Electrophysiological Insight: 3 physical orthogonal dipoles span the cardiac electrical space; 1 lead requires discrete latent regularizers.